Bjarne R. Bartlett

bjarne@hawaii.edu

EDUCATION

B.S., Gettysburg College *2008-2012*

- Major: Biology
- Kappa Delta Pi Honor Society, Secretary

Ph.D. Candidate, The University of Hawaiʻi at Mānoa *2017-present*

- Major: Molecular Biosciences and Bioengineering *A.B.D 05/2021*

Anticipated graduation date: 08/2022

EMPLOYMENT

Data Science Graduate Fellowship, The University of Hawaiʻi at Mānoa *2017-present*
Mentors: Drs. Mikey Kantar, Jon-Paul Bingham, Monica Stitt-Bergh, Amy Hubbard, Gernot Presting, Youping Deng, Sean Clevaland, Gwen Jacobs

- Developed prognostic signatures for patients with colorectal cancer using using a custom R program and gene expression profiles from The Cancer Genome Atlas – this project published in the *International Journal of Computational Biology and Drug Design*
- Explored the miRNA expression profiles of inflammatory tumors using R and data from The Cancer Genome Atlas to find an immune cell profile favorable to treatment by checkpoint blockade immunotherapy
- Identified data from Altmetrics, a research attention metric, as a less-biased measure of research productivity than traditional metrics using Python and Jupyter Notebooks, leading to a publications in *Scientometrics* and *Nature Index*
- Performed a genome and transcriptome assembly of *Macadamia tetraphylla*, a crop wild relative of *Macadamia integrifolia*, to investigate potential mechanisms for disease resistant crops using next generation sequencing tools
- Developed a novel fingerprinting method for *Colocasia esculenta* using genetic regions capable of distinguishing between Hawaiian and non-Hawaiian varieties

Consultant, Personal Genome Diagnostics *2015-2017*
Mentors: Drs. Luis Diaz, Victor Velculescu, Lisa Kann

- Initiated and completed the validation of a new peripheral blood test to detect clinically actionable genetic alterations in non-small cell lung cancer patients
- Followed CLIA approved protocols to prepare clinical samples for a variety next generation sequencing assays

Research Specialist, Johns Hopkins School of Medicine, Department of Oncology *2012-2017*
Mentors: Drs. Bert Vogelstein, Ken Kinzler, and Luis Diaz

- Designed and coordinated a phase 2 clinical trial, testing the efficacy of anti-programmed-death-1 (anti-PD-1) therapy in mismatch repair deficient tumors leading to a publication in the *New England Journal of Medicine*
- Identified and screened over 500 patients for ongoing enrollment on the anti-pd-1 clinical trial
- Developed a bioinformatics pipeline in UNIX using Python and NetMHC 3.4 to predict neoepitope burden in tumors of immunotherapy patients

- Evaluated the sensitivity and specificity of circulating tumor DNA (ctDNA) in peripheral blood as a non-invasive diagnostic test for cancer by collecting blood and isolating ctDNA in 640 patients leading to a publication in *Science Translational Medicine*
- Developed a phase 3 clinical trial protocol to test DNA from the Papanicolaou test as a non-invasive diagnostic for ovarian or endometrial cancers and optimized collection methods in the protocol to increase the sensitivity of the test
- Performed correlative studies and data analysis to corroborate a clinical trial showing the efficacy of combining gemcitabine, taxotere, and xeloda (GTX) with cisplatin for metastatic pancreatic cancer

MANUSCRIPTS IN PREPARATION

- **Bartlett BR**, Kantar MB, Bingham JP. "A Stand-Alone Data Science Primer for Undergraduate Biomedical Research Students." In preparation.
- **Bartlett BR**, Zamora CM**^**, Kantar MB. "A Novel Transcriptome Assembly for *Macadamia tetraphylla*." In preparation.
- **Bartlett BR**, Zamora CM**^**, Kantar MB. "A Genetic Fingerprint for *Colocasia esculenta*." In preparation.
- **Bartlett BR**, Zamora CM**^**, Bingham JP, Hubbard A, Runck B, Kantar MB. "Journal influence on the short and long term impact of science." In preparation.

PUBLICATIONS

h-index: 13 | **i0-index:** 13 Google Scholar
h-index: 13 Web of Science
ResearchGate: 29.28 ResearchGate

- **Bartlett B**, Gao Z, Schukking M, Menor M, Khadka VS, Fabbri M, Deng Y. The miRNA Profile of Inflammatory Colorectal Tumors Identify TGF-β as a Companion Target for Checkpoint Blockade Immunotherapy. *Frontiers in Cell and Developmental Biology*. 2021 Oct: 2757.
  conception/design (100%)
  acquisition/analysis of data (50%)
  creation of new software (100%)
  drafted/revised the manuscript (50%)

- Fortin J, **Bartlett B**, Kantar M, Tseng M, Mehrabi Z. "Digital technology helps remove gender bias in academia." *Scientometrics*. 2021 May;126(5):4073-81.
  conception/design (10%)
  acquisition/analysis of data (90%)
  creation of new software (50%)
  drafted/revised the manuscript (50%)
- **Bartlett B**, Fortin J, Kantar M, Tseng M, Mehrabi Z. "How altmetrics could help level the playing field for women in STEM." *Nature Index*. 2021 March.
  conception/design (90%)
  acquisition/analysis of data (90%)
  creation of new software (n/a)
  drafted/revised the manuscript (50%)
- **Bartlett B**, Zhu Y, Menor M, Khadka VS, Zhang J, Zheng J, Jiang B, Deng Y. "Development of a RNA-Seq based prognostic signature for colon cancer." *International Journal of Computational Biology and Drug Design*. 2020;13(5-6):488-503.

conception/design (90%)
acquisition/analysis of data (90%)
creation of new software (90%)
drafted/revised the manuscript (90%)

- Georgiadis A, Durham JN, Keefer LA, **Bartlett BR**, Zielonka M, Murphy D, White JR, Lu S, Verner EL, Ruan F, Riley D, et al. "Noninvasive detection of microsatellite instability and high tumor mutation burden in cancer patients treated with PD-1 blockade." *Clinical Cancer Research*. 2019 Dec 1;25(23):7024-34.
  - conception/design (10%)
  - acquisition/analysis of data (90%)
  - creation of new software (50%)
  - drafted/revised the manuscript (50%)
- Smith KN, Llosa NJ, Cottrell TR, Siegel N, Fan H, Suri P, Chan HY, Guo H, Oke T, Awan AH, Verde F, Danilova L, Anagnostou V, Tam AJ, Luber BS, **Bartlett BR**, et al.. "Persistent mutant oncogene specific T cells in two patients benefitting from anti-PD-1." *Journal for immunotherapy of cancer*. 2019 Dec;7(1):40.
  - conception/design (10%)
  - acquisition/analysis of data (10%)
  - creation of new software (50%)
  - drafted/revised the manuscript (10%)
- Llosa NJ, Luber B, Tam AJ, Smith KN, Siegel N, Awan AH, Fan H, Oke T, Zhang J, Domingue J, Engle EL, Roberts CA, **Bartlett BR**, et al.. "Intratumoral Adaptive Immunosuppression and Type 17 Immunity in Mismatch Repair Proficient Colorectal Tumors." *Clinical Cancer Research*. 2019 May 6.
  - conception/design (10%)
  - acquisition/analysis of data (50%)
  - creation of new software (50%)
  - drafted/revised the manuscript (10%)
- Mandal R, Samstein RM, Lee KW, Havel JJ, Wang H, Krishna C, Sabio EY, Makarov V, Kuo F, Blecua P, Ramaswamy AT, Durham JN, **Bartlett B**, et al. "Genetic diversity of tumors with mismatch repair deficiency influences anti–PD-1 immunotherapy response." *Science*. 2019 May 3;364(6439):485-91.*****
  - conception/design (20%)
  - acquisition/analysis of data (30%)
  - creation of new software (50%)
  - drafted/revised the manuscript (20%)
- Le DT, Durham JN, Smith KN, Wang H, **Bartlett BR**, Aulakh LK, Lu S, Kemberling H, Wilt C, Luber BS, Wong F, et al. "Mismatch repair deficiency predicts response of solid tumors to PD-1 blockade." *Science*. 2017 Jul 28;357(6349):409-13. *****
  - conception/design (20%)
  - acquisition/analysis of data (30%)
  - creation of new software (90%)
  - drafted/revised the manuscript (20%)
- Parpart-Li S, **Bartlett B**, Popoli M, Adleff V, Tucker L, Steinberg R, Georgiadis A, Phallen J, Brahmer JR, Azad NA, Browner I, Laheru D, Velculescu V, Sausen M, Diaz LA Jr. "The effect of preservative and temperature on the analysis of circulating tumor DNA." *Clinical Cancer Research.* 2016; 23 (10): 2471-2477.*****
  - conception/design (50%)
  - acquisition/analysis of data (50%)

creation of new software (n/a)
drafted/revised the manuscript (50%)

- Le DT, Uram JN, Wang H, **Bartlett BR**, Kemberling H, Eyring AD, Skora AD, Luber BS, Azad NS, Laheru D, Biedrzycki B, et al. "PD-1 blockade in tumors with mismatch-repair deficiency." *New England Journal of Medicine*. 2015; 372(26):2509-2520. *****
  conception/design (10%)
  acquisition/analysis of data (30%)
  creation of new software (90%)
  drafted/revised the manuscript (10%)
- Bettegowda C, Sausen M, Leary RJ, Kinde I, Wang Y, Agrawal N, **Bartlett BR**, Wang H, Luber B, Alani RM, Antonarakis ES, et al. "Detection of circulating tumor DNA in early- and late-stage human malignancies." *Science Translational Medicine*. 2014; 6(224):224ra24. *****
  conception/design (5%)
  acquisition/analysis of data (50%)
  creation of new software (50%)
  drafted/revised the manuscript (10%)

*Cited more than 100 times
^Undergraduate Student Author

POSTERS AND PRESENTATIONS

- **Bartlett BR**. "Using RNA-Seq to Predict Prognosis for Colon Cancer Patients." International Conference on Intelligent Biology and Medicine (2019). Selected for oral presentation.
- Yu Z, Zhu Y, Ai J, **Bartlett B**, Zhang J, Jiang B, Deng Y. "Development of predictive models to distinguish metals from non-metal toxicants, and individual metal from one another." International Conference on Intelligent Biology and Medicine (2019). Selected for oral presentation.
- **Bartlett BR.** "The miRNA Expression Profile of Inflammatory Tumors Reveals a Unique Immune Cell Profile and Potential Companion Targets for Checkpoint Blockade Immunotherapy." AACR-JCA Joint Conference (2019). Selected for poster.
- **Bartlett B**, Khadka V, Menor M, Deng Y. "The miRNA expression profile of inflammatory tumors reveals a unique immune cell profile and potential companion targets for checkpoint blockade immunotherapy." AACR (2019). Selected for poster.
- Georgiadis A, Durham JN, Keefer L, **Bartlett BR**, Zielonka M, Murphy D, White JR, Lu S, Verner E, Ruan F, Riley D. "Analysis of cell-free plasma DNA to identify tumors with microsatellite instability and exceptionally high tumor mutation burden in patients treated with PD-1 blockade." European Journal Of Cancer (2018).
- **Bartlett BR**. "Developing a solid-tumor screening program checkpoint blockade immunotherapy." The University of Hawaiʻi (2017). Invited speaker.
- Parpart-Li ST, **Bartlett B**, Popoli M, Adleff V, Brahmer J, Azad N, Bonerigo S, Browner I, Ryan A, Velculescu V, Sausen M. "Optimized plasma collection procedures for liquid biopsy analyses in cancer." AACR (2016).
- Le DT, Uram JN, Wang H, **Bartlett B**, Kemberling H, Eyring A, Azad NS, Laheru D, Donehower RC, Crocenzi TS, Goldberg RM. "Programmed death-1 blockade in mismatch repair deficient colorectal cancer." Journal of Clinical Oncology (ASCO Meeting Abstracts) 2016
- Diaz LA, Uram JN, Wang H, **Bartlett B**, Kemberling H, Eyring A, Azad NS, Dauses T, Laheru D, Lee JJ, Crocenzi TS. "Programmed death-1 blockade in mismatch repair deficient cancer independent of tumor histology." Journal of Clinical Oncology (ASCO Meeting Abstracts) 2016
- Le DT, Uram JN, Wang H, Kemberling H, Eyring A, **Bartlett B**, Goldberg R, Crocenzi TS, Fisher GA, Lee JJ, Greten TF, Laheru DA, Azad NS, Donehower RC, Luber B, Koshiji M, Eshleman JR, Anders RA, Vogelstein B, Diaz LA. "PD-1 blockade in mismatch repair deficient non-colorectal

gastrointestinal cancers." Journal of Clinical Oncology (ASCO Meeting Abstracts). 2016;34;(abstracts 195).
- Parpart-Li S, **Bartlett B**, Brahmer J, Azad N, Bonerigo S, Browner I, Ryan A, Velculescu V, Sausen M, Diaz L. "Optimized plasma collection procedures for cell-free DNA analysis in advances malignancies." Circulating nucleic acids in plasma and serum (CNAPS) IX congress (2015). Selected for oral presentation.
- Le DT, Uram JN, Wang H, **Bartlett B**, Kemberling H, Eyring A, Skora A, Azad NS, Laheru DA, Donehower RC, Luber B, Crocenzi TS, Fisher GA, Duffy SM, Lee JJ, Koshiji M, Eshleman JR, Anders RA, Vogelstein B, Diaz LA. "PD-1 blockade in tumors with mismatch repair deficiency." Journal of Clinical Oncology (ASCO Meeting Abstracts). 2015;33(15);(abstracts LBA100).
- Wang JS, Sausen M, Parpart-Li S, Murphy DM, Velculescu VE, Wood LD, Solt-Linville S, Sugar E, **Bartlett B**, Blair C, Dauses T, Jaffee EM, Hruban RH, Laheru D, Diaz LA. "Circulating tumor DNA (ctDNA) as a prognostic marker for recurrence in resected pancreas cancer." Journal of Clinical Oncology (ASCO Meeting Abstracts). 2015;33(15):11025.
- Bettegowda C, Sausen M, Leary R, Kinde I, Agrawal N, **Bartlett B**, Wang H, Luber B, Kinzler K, Vogelstein B, Papadopoulos N, Diaz L. "Abstract 5606: Detection of circulating tumor DNA in early and late stage human malignancies." *Cancer Research*. 2014; 74(19 Supplement):5606.
- **Bartlett BR**, Keane B, Fong P. "The Effect of Venlofaxin on the Burrowing Speed of *Corbicula fluminea..*" Gettysburg College capstone poster presentation. 2011. (Advisor Dr. Peter Fong).

HONORS AND AWARDS

CI-TRACS Data Science Fellow, University of Hawaiʻi Cyberinfrastructure *2021*
- Selected to be a fellow with the Hawaiʻi Data Science Institute as part of a $1 million National Science Foundation grant to increase student cyberinfrastructure skills in climate science.
- Article: $1M supports resilience to climate change through cyberinfrastructure training

Featured Researcher, University of Hawaiʻi News *2021*
- Article: New metrics could reduce gender gap in STEM fields

Merit-Based Scholarship, MBBE *2020*
- Received a $1,000.00, merit-based scholarship for academic productivity as a graduate student

Recognition for 50 Years of Broadcasting, Honolulu City Council *2019*
- KTUH Honolulu recognized for 50 years of college radio broadcasting

Scholar in Training Award Recipient, AACR-JCA Joint Conference *2019*
- Received an $800.00, merit-based award from AACR to present Master's Degree thesis project: "The miRNA Expression Profile of Inflammatory Tumors Reveals a Unique Immune Cell Profile and Potential Companion Targets for Checkpoint Blockade Immunotherapy."

Travel Award Recipient, The International Conference on Intelligent Biology & Medicine *2019*
- Received a $600.00, merit-based award from ICIBM to present master's research project: "Using RNA-Seq to Predict Prognosis for Colon Cancer Patients"

Best College Radio Website, College Broadcasters International *2019*
- As Web Director recognized as a finalist for best college radio website: KTUH.org

CONFERENCES/PROFESSIONAL SOCIETIES

Member, Institute of Electrical and Electronics Engineers *2017-present*
Associate Member, The American Association for Cancer Research *2016-present*
Member, American Association for the Advancement of Science *2016-present*

Meeting Volunteer, American Association for the Advancement of Science (Seattle, WA) *2020*

- Volunteered as a staff member and educator for AAAS Family Science Days, a free public science event offering child-friendly activities.

Meeting Coordinator, International Conference of Translational Medicine (Honolulu, HI) *2019*

- Collaborated with scientists from Asia to organize a 2-day, international meeting on translational medicine

Meeting Volunteer, Engineering in Medicine and Biology Conference (Honolulu, HI) *2018*
Host: Institute of Electrical and Electronics Engineers

- In addition to the meeting, attended a special workshop: "Writing a patent application for biomedical technologies. How to do it, what is important, how to write claims and where to file?"

Secretary General, Circulating Nucleic Acids in Plasma and Serum Meeting (CNAPS VIII) *2013*

- Organized an international meeting for over 250 scientists
- Recruited 20 internationally recognized speakers to lecture and chair 5 sessions
- Identified promising abstracts and forwarded them to session chairs
- Managed a $150,000 meeting budget

<u>TEACHING</u>

Research Mentor, University of Hawaiʻi, Research Experience for Undergraduates *2021*
Mentee: Carter Zamora from Gettysburg College

- Performed basic computational biology and analysis in R and using the University of Hawaiʻi High Performance Computing Cluster.

Graduate Teaching Assistant, University of Hawaiʻi, INBRE Program *2020-2021*

- Taught Introductory Bioinformatics Seminar

Graduate Teaching Assistant, University of Hawaiʻi, Department of Microbiology *2018*

- Taught MICR 461L, a lab-based immunology course

Student Research Advisor, Johns Hopkins School of Medicine, Department of Oncology *2013-2017*

- Laboratory of Bert Vogelstein, Ken Kinzler, and Luis Diaz
    - Supervised 3 summer rotation students working independently on research projects related to ongoing clinical trials

Peer Learning Associate, Gettysburg College *2010-2012*

- The Joy of Science: Returning Discovery to Science Learning, Educational Psychology, The Social Foundations of Education
- Assisted faculty in teaching 2 core education courses
- Designed and led outside class sessions for a seminar on STEM education

<u>FUNDING RECEIVED</u>

SEED IDEAS Diversity Grant, Kantar Lab *2020*

- Topic: How altmetrics could help level the playing field for women in STEM
- Recipient of a SEED IDEAS grant to bring a visiting scholar to discuss *Digital technology helps remove gender bias in academia*

SEED IDEAS Diversity Grant, KTUH Honolulu *2019*

- Topic: MALAMA Aquaponics Program
- Recipient of a SEED IDEAS grant to produce the first podcasts from KTUH

AAAS Science Outreach Grant, Kantar Lab *2018*

- Topic: The Power of Infographics
- Recipient of an AAAS communicating science grant with Dr. Michael Kantar

COMPUTER/LAB SKILLS

Programming Languages: R, Python, Java, HTML, JavaScript, CSS

Bioinformatic Skills: Processing and analyzing next generation sequencing data; somatic cancer mutation analysis, cancer neoantigen analysis; microbiome analysis; cancer biomarker discovery; genome assembly; DNA microsatellite instability analysis

Lab Skills: Preparation and storage of clinical samples including blood, saliva, fresh tumor tissue, lymphocytes, and fixed tissue; DNA extraction from whole blood and blood plasma; dual DNA extractions from formalin fixed tissue for comparative tumor/normal next generation sequencing of cancers; blood fractionation; Ficoll separations; fermentation biochemistry; high-performance liquid chromatography

COMMUNITY SERVICE/LEADERSHIP

General Manager, KTUH Honolulu *2018-2021*

- Promoted from Web Director & Program Director
- Successfully tripled annual KTUH fundraising goals for 2 years as station director
- Managed a student-fee funded budget of $111,000
- Founded and managed an endowment fund for KTUH of $50,000
- Founded and managed an endowed scholarship fund for KTUH of $50,000
- Created a brand-new website which was selected as the finalist at College Broadcasters International
- Hosted 50th anniversary concert series with finale concert headlined by Richard Thompson
- Produced and curated an exhibit at the Hamilton Library
- Started the first KTUH-hosted podcast channel (as Program Director) made available on SoundCloud, iTunes and other platforms

Volunteer, GENE-ius Day Summer Program *2019*

- Coordinated *Speaking Science*, a summer program for local high school students in Hawaiʻi.
- Students worked in teams to produce podcast episodes about research at the University of Hawaiʻi at Mānoa

Team Captain, Swim Across America Baltimore *2012-2015*

- Grew the Johns Hopkins Kimmel Cancer Center team from 5 to 18 participants
- Increased fundraising from $2000.00 in 2012 to over $12,000.00 in 2015

Co-director Planning Committee, Johns Hopkins Oncology Survivor Clinic *2012-2015*

- Designed and implemented the first retreat at Johns Hopkins for patients with metastatic colon cancer and their spouses
- Hosted retreats on an annual basis and expanded to patients with pancreatic cancer because of the success of the initial evens
- Recruited experts in oncology, nutrition, complementary and alternative medicine, and finance

Runner, Baltimore Running Festival *2014*

- Ran on the Pancreatic Cancer Action Network's relay team for the Baltimore marathon

Tutor, El Centro Community Center (Gettysburg, PA) *2010-2012*

- Tutored reading and English to students from local Gettysburg schools

College Life Technician, Gettysburg College *2009-2012*

- Contact event sponsors to ensure timely setup and correct equipment has been requested
- Operate lighting and sound equipment at The Attic, The Junction, and The Ballroom
- Performances included formal dances, professional bands, speakers, soloists, and student groups

Residence Coordinator, Gettysburg College *2009-2012*

- One of two junior students selected to be residence coordinators
- Supervised four student staff members in Residence Life
- Directed RISE, an organization that implemented substance free programming on campus

REFERENCES

Michael B. Kantar 2019-present
3190 Maile Way, Room 102, Honolulu, Hawaii 96822
(808) 956-2162
mbkantar@hawaii.edu
Direct supervisor and doctoral committee chair at the University of Hawaiʻi.

Jon-Paul Bingham *2017-present*
1955 East West Rd. Agricultural Sciences 218, Honolulu, HI 96822
(808) 956-4864
jbingham@hawaii.edu
Teaching supervisor and graduate chair at the University of Hawaiʻi.

Luis A. Diaz Jr. MD *2012-2017*
1275 York Avenue, New York, NY 10065
(212) 639-2000
ldiaz@mskcc.org
Direct supervisor and research mentor at Johns Hopkins University.